

William Turchin

Colorado Springs, CO | (425) 414-9745 | wturchin@gmail.com | linkedin.com/in/william-turchin

Education

Colorado School of Mines Golden, CO
Combined BS/MS in Computer Science, GPA: 3.898 *Expected May 2027 (BS), May 2028 (MS)*

- **Relevant Coursework:** Systems Programming, Databases, Probability & Statistics, Advanced Algorithms
- **Activities:** Formula SAE, Resident Advisor

Experience

Software Engineering Intern, Platform May 2026 – Aug. 2026
Ibotta *Denver, CO*

- Migrated Databricks warehouses to serverless, eliminating 30-second to 5-minute cold starts.
- Built a 7-spec Playwright E2E suite and promoted it to a required CI gate after validating under 5% flake.
- Built an agentic repo-context system across 1,700 repositories; improved coding-agent performance 2%.
- Built an AWS Bedrock AgentCore backend and GitHub webhook to maintain repository context.

Software Engineering Intern Jan. 2025 – Aug. 2025
Raven Biosystems *Seattle, WA*

- Designed camera and computer vision systems for the patented RavenScope™ digital microscope.
- Built a 64 MP Raspberry Pi imaging system with automated focusing, cropping, and sample-region detection.
- Automated 20 μ m-resolution slide analysis in under 5 minutes with Java-based ImageJ and Python workflows.

Data Systems Lead Aug. 2024 – Present
Mines Formula Team *Golden, CO*

- Led telemetry and visualization for a Formula-style racecar; mentored four junior team members.
- Processed data from 10+ CAN, GPS, acceleration, wheel-speed, and temperature sensors at up to 20 Hz.
- Built Rerun visualization and state-estimation tools to diagnose vehicle behavior and bottlenecks.
- Built a CAN driver display for telemetry and engine-failure warnings with under 20 ms latency.

Projects

Botta Daily Spin — Ibotta Hackathon Winner | *React, TypeScript, CSS*

- Won People's Choice among 29 projects as an intern team competing against full-time employees.
- Built a React prize wheel with weighted results, easing, reveals, and rare-reward effects for daily engagement.

Self-Hosted Platform | *Linux, Docker Compose, GitHub Actions, Borg, Grafana*

- Operated a multi-service environment on a Dell PowerEdge R640 running Ubuntu and Docker Compose.
- Configured a self-hosted GitHub Actions runner with automated service deployment after repository updates.
- Implemented local/remote Borg backups, health monitoring, and Grafana infrastructure observability.

Formula SAE Telemetry & Observability Platform | *InfluxDB, Grafana, Docker, Linux, Python*

- Deployed Dockerized InfluxDB and Grafana telemetry for 10+ vehicle sensors at 20 Hz.
- Built real-time dashboards for CAN, GPS, acceleration, wheel speed, and temperature data.
- Standardized time-series schemas and ingestion across electronic control units and sensor frequencies.
- Migrated telemetry to a Dell PowerEdge server with Docker Compose, backups, and health monitoring.
- Led telemetry development; mentored four teammates on architecture, Git, and deployments.

AI Classical Music Generator | *Python, PyTorch*

- Trained a PyTorch-based sequence model on classical music to generate melodies.
- Developed a custom dataset to convert 35 MIDI files into 1,000 tokenized input sequences for training.
- Generated polyphonic sequences with recognizable motifs after 50 epochs of training.

Skills

Languages: C/C++, Python, TypeScript, Java, SQL, Assembly

Libraries & Data: PyTorch, Pandas, NumPy, Scikit-learn

Cloud & Infrastructure: AWS Bedrock, Docker, GitHub Actions, Linux, Terraform, Databricks, InfluxDB, Grafana

Frameworks & Systems: React, Playwright, CMake, Protobuf, CAN bus, Git

AI Systems: LLM APIs, AWS Bedrock, MCP, LLM Evaluation, Agentic Workflows